Exercitii

L·U

Crout

ex.: $b = \begin{bmatrix} 7 \\ 9 \\ 1 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

- $A = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 1 & 2 \\ -2 & 0 & 1 \end{bmatrix}$; $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$; $U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$

$$L \cdot U = \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12}+l_{22} & l_{21}u_{13}+l_{22}u_{23} \\ l_{31} & l_{31}u_{12}+l_{32} & l_{31}u_{13}+l_{32}u_{23}+l_{33} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 3 & 0 \\ 1 & 1 & 2 \\ -2 & 0 & 1 \end{bmatrix}}_{A}$$

$l_{11} = 1$; $l_{11} \cdot u_{12} = 3 \Rightarrow u_{12} = 3$; $l_{11} u_{13} = 0 \Rightarrow u_{13} = 0$

$l_{21} = 1$; $1 \cdot 3 + l_{22} = 1 \Rightarrow l_{22} = -2$; $1 \cdot 0 + (-2) \cdot u_{23} = 2 \Rightarrow u_{23} = -1$

$l_{31} = -2$; $-2 \cdot 3 + l_{32} = 0 \Rightarrow l_{32} = 6$; $-2 \cdot 0 + 6 \cdot (-1) + l_{33} = 1 \Rightarrow l_{33} = 7$

$$\Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ -2 & 6 & 7 \end{bmatrix} ; \quad U = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$A = L \cdot U$; $A \cdot x = b \Rightarrow L \cdot \underbrace{U \cdot x}_{\text{mut}} = b$

$U \cdot x = y$ SIS ... $\Rightarrow y = [\,;\,]$ $\quad y \neq b$

$L \cdot y = b$ SST ... $\Rightarrow b = [\,;\,]$

Doolittle

ex: $b = \begin{bmatrix} 7 \\ 9 \\ 1 \end{bmatrix} \Rightarrow x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ ✓

- $A = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 1 & 2 \\ -2 & 0 & 1 \end{bmatrix}$; $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$; $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 1 & 2 \\ -2 & 0 & 1 \end{bmatrix} = L \cdot U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

L_1: $\boxed{u_{11} = 1}$; $\boxed{u_{12} = 3}$; $\boxed{u_{13} = 0}$

C_1: $1 \cdot l_{21} = 1 \Rightarrow \boxed{l_{21} = 1}$; $1 \cdot l_{31} = -2 \Rightarrow \boxed{l_{31} = -2}$

L_2: $3 \cdot 1 + u_{22} = 1 \Rightarrow \boxed{u_{22} = -2}$; $1 \cdot 0 + u_{23} = 2 \Rightarrow \boxed{u_{23} = 2}$

C_2: $-2 \cdot 3 + l_{32} \cdot (-2) = 0 \Rightarrow \boxed{l_{32} = -3}$

L_3: $-2 \cdot 0 + \underbrace{(-3) \cdot 2}_{-6} + u_{33} = 1 \Rightarrow \boxed{u_{33} = 7}$

$$\Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & -3 & 1 \end{bmatrix}; \quad U = \begin{bmatrix} 1 & 3 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & 7 \end{bmatrix}$$

Crout 3 diag

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ 0 & l_{32} & l_{33} \end{bmatrix}; \quad U = \begin{bmatrix} 1 & u_{12} & 0 \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$L \cdot U = \begin{bmatrix} l_{11} & l_{11} \cdot u_{12} & 0 \\ l_{21} & l_{21} \cdot u_{12} + l_{22} & l_{22} \cdot u_{23} \\ 0 & l_{32} & l_{32} \cdot u_{23} + l_{33} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 3 & 0 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}}_{A}$$

L_1: $\boxed{l_{11} = 1}$; $1 \cdot u_{12} = 3 \Rightarrow \boxed{u_{12} = 3}$

C_1: $\boxed{l_{21} = 1}$

L_2: $1 \cdot 3 + l_{22} = 1 \Rightarrow \boxed{l_{22} = -2}$; $-2 \cdot u_{23} = 2 \Rightarrow \boxed{u_{23} = -1}$

C_2: $\boxed{l_{32} = 0}$

L_3: $0 \cdot u(-1) + l_{33} = 1 \Rightarrow \boxed{l_{33} = 1}$

$$\Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad U = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

Cholesky : A < simetrică, pozitiv definită (Cf. Reg. Sylvester → Toți minorii princ. > 0)

$A = L \cdot L^T$; L - inf Δ ($\Rightarrow L^T$ - sup Δ)

- $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 5 \end{bmatrix}$

Obs: $l_{ij} = l_{ji}^T$

$L \cdot L^T = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \cdot \begin{bmatrix} l_{11} & l_{12} & l_{13} \\ 0 & l_{22} & l_{23} \\ 0 & 0 & l_{33} \end{bmatrix} = A \Leftrightarrow$

$\Leftrightarrow \begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 5 \end{bmatrix} = \begin{bmatrix} l_{11}^2 & l_{11}l_{12} & l_{11}l_{13} \\ l_{21}l_{11} & l_{21}l_{12}+l_{22}^2 & l_{21}l_{31}+l_{22}l_{23} \\ l_{31}l_{11} & l_{31}l_{12}+l_{32}l_{22} & l_{31}l_{13}+l_{32}l_{23}+l_{33}^2 \end{bmatrix}$

Analizăm Δ sup SAU Δ inf : (+ D.P.)

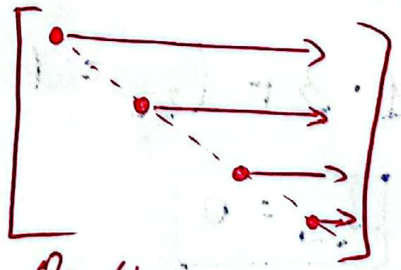
L_1: $l_{11}^2 = 3 \Rightarrow l_{11} = \sqrt{3}$; $\sqrt{3} \cdot l_{12} = -1 \Rightarrow l_{12} = -\frac{1}{\sqrt{3}}$

$\sqrt{3} \cdot l_{13} = 1 \Rightarrow l_{13} = \frac{1}{\sqrt{3}}$

L_2: $l_{21} \cdot \left(\frac{-1}{\sqrt{3}}\right)^2 + l_{22}^2 = 3 \Rightarrow l_{22} = \sqrt{\frac{8}{3}}$

$-\frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} + \sqrt{\frac{8}{3}} \cdot l_{23} = 2 \Rightarrow l_{23} = \sqrt{\frac{49}{24}}$

L_3: $\frac{1}{3} + \frac{49}{24} + l_{33}^2 = 5 \Leftrightarrow l_{33} = \sqrt{\frac{71}{24}}$



Ordinea de scriere a ec.

$\Rightarrow L = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ -1/\sqrt{3} & \sqrt{8/3} & 0 \\ 1/\sqrt{3} & \sqrt{49/24} & \sqrt{71/24} \end{bmatrix}$

LDL^T: A < simetrică, pozitiv definită

$A = L \cdot D \cdot L^T$ < L: inf. Δ cu 1 pe D.P.; D: mat. diag. cu el. de pe D.P. > 0

$$\cdot A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 5 \end{bmatrix}$$

$$L \cdot D \cdot L^T = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}}_{L} \cdot \underbrace{\begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix}}_{D} \cdot \underbrace{\begin{bmatrix} 1 & l_{21} & l_{31} \\ 0 & 1 & l_{32} \\ 0 & 0 & 1 \end{bmatrix}}_{L^T}$$

$$= \underbrace{\begin{bmatrix} d_{11} & 0 & 0 \\ l_{21} d_{11} & d_{22} & 0 \\ l_{31} d_{11} & l_{32} d_{22} & d_{33} \end{bmatrix}}_{L \cdot D} \cdot \underbrace{\begin{bmatrix} 1 & l_{21} & l_{31} \\ 0 & 1 & l_{32} \\ 0 & 0 & 1 \end{bmatrix}}_{L^T} =$$

$$= \underbrace{\begin{bmatrix} d_{11} & d_{11} l_{21} & d_{11} l_{31} \\ l_{21} d_{11} & l_{21}^2 d_{11} + d_{22} & l_{21} d_{11} l_{31} + l_{32} d_{22} \\ l_{31} d_{11} & l_{31} d_{11} l_{21} + l_{32} d_{22} & l_{31}^2 d_{11} + l_{32}^2 d_{22} + d_{33} \end{bmatrix}}_{L \cdot D \cdot L^T \text{ (sim.)}} = A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 5 \end{bmatrix}$$

$\Rightarrow$ ég. Δ sup (sau Δ inf) $\Rightarrow$

$$\Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 \\ \frac{1}{3} & \frac{7}{8} & 1 \end{bmatrix}; \quad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2{,}667 & 0 \\ 0 & 0 & 2{,}625 \end{bmatrix}$$

Givens

(Q · R)

$$G = \begin{bmatrix} \cos & -\sin \\ \sin & \cos \end{bmatrix}$$

- $A = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix}$;

$$\begin{cases} \cos * = \dfrac{x(1)}{\sqrt{x^2(1)+x^2(2)}} \\ \sin * = -\dfrac{x(2)}{\sqrt{x^2(1)+x^2(2)}} \end{cases}$$

$A^{(0)} = A$

→ Anulăm $A^{(0)}(2,1) = 0$:

$$c_{12} = \frac{3}{\sqrt{3^2+0^2}} = 1 \; ; \; s_{12} = \frac{-0}{\sqrt{3^2+0^2}} = 0 \Rightarrow$$

G_{12}^* - submat. a mat G_{12}:

$$\Rightarrow G_{12}^* = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} ; \; G_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

se selectează L_1 și L_2 din matricea $A^{(0)}$

$$G_{12}^* \cdot \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \end{bmatrix} \begin{matrix} L_1 \\ L_2 \end{matrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \end{bmatrix} \Rightarrow$$

submatrice a mat. $A^{(0)}$

$$\Rightarrow A^{(1)} = G_{12} \cdot A^{(0)} = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix}$$

→ Anulăm $A^{(1)}(3,1) = 4$:

$$c_{13} = \frac{3}{\sqrt{3^2+4^2}} = \frac{3}{5}$$

$$s_{13} = \frac{-4}{\sqrt{3^2+4^2}} = -\frac{4}{5}$$

$$\Rightarrow G_{13}^* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$\Rightarrow A^{(2)} = G_{13} \cdot \underbrace{A^{(1)}}_{G_{12} \cdot A^{(0)}} = G_{13} \cdot G_{12} \cdot A^{(0)} = \begin{bmatrix} 5 & 2 & 5 \\ 0 & 3 & 5 \\ 0 & 4 & 0 \end{bmatrix}$$

→ Anulăm $A^{(2)}(3,2) = 4$:

$$c_{23} = \frac{3}{\sqrt{3^2+4^2}}$$

$$s_{23} = \frac{-4}{\sqrt{3^2+4^2}}$$

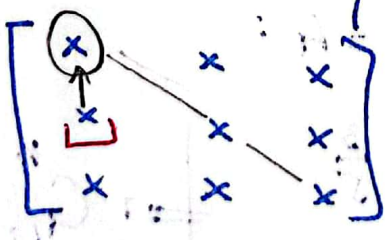
$$\Longrightarrow G_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3/5 & 4/5 \\ 0 & -4/5 & 3/5 \end{bmatrix}.$$

$$\Longrightarrow \underline{A^{(3)}} = G_{23} \cdot A^{(2)} = \overbrace{G_{23} \cdot G_{13} \cdot G_{12}}^{G} \cdot A^{(1)} \Longleftrightarrow$$

$$\Longleftrightarrow A^{(3)} = \begin{bmatrix} 5 & 2 & 5 \\ 0 & 5 & 3 \\ 0 & 0 & -4 \end{bmatrix} = \underline{R} \ (\text{sup } \Delta)$$

$\underline{Q = G^T}$, unde $G = G_{23} \cdot G_{13} \cdot G_{12}$.

Obs.: Pentru a introduce un zero, ne folosim de elementul de pe D.P. din acea coloană:



Householder

$A = Q \cdot R$ (Q - ortogonală, R - sup$\triangle$)

$A \in \mathbb{R}^{m \times n}$; $Q \in \mathbb{R}^{m \times m}$; $R \in \mathbb{R}^{m \times n}$

m = nr. de linii

- $A = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix}_{3 \times 3}$

Pas $\boxed{p = 1}$: $A_1 \xleftarrow{not} A$

$A \in \mathbb{R}^{3 \times 3}$; $m = 3$; $n = 3$

$H = H_{min(m-1,n)} \cdots H_1$

$\Rightarrow \boxed{H = H_2 \cdot H_1} \Rightarrow$ (2 pași)

a_1 - vect. col (1) din mat. A_1

$a_1 = \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$;

(*) $v_{pp} = a_{pp} + \sigma_p$

$\sigma_p = semn(a_{pp}) \cdot norm(A[p:m, p])$ — vect. col "p"

(**) $v_{ip} = a_{ip}$, $i = p+1 : m$

(*) + (**) $\Rightarrow H_p = I_m - 2 \cdot \frac{v_p \cdot v_p^T}{\|v_p\|_2^2}$

$H = H_{min(m-1,m)} \cdots H_p \cdots H_2 \cdot H_1$

$\sigma_1 = semn(\underbrace{a_{11}}_{3}) \cdot \|a_1\| = +\sqrt{3^2 + 0^2 + 4^2} = 5$

$v_1 = \begin{bmatrix} (*) a_{11} + \sigma_1 \\ (**) a_{21} \\ (**) a_{31} \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 4 \end{bmatrix} \Rightarrow H_1 = I_3 - 2 \cdot \frac{v_1 \cdot v_1^T}{\|v_1\|^2} \Leftrightarrow$

$\Leftrightarrow H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \cdot \frac{1}{80} \cdot \begin{bmatrix} 8 \\ 0 \\ 4 \end{bmatrix} \cdot [8 \ 0 \ 4] \Leftrightarrow$

$\Leftrightarrow H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{40} \cdot \begin{bmatrix} 64 & 0 & 32 \\ 0 & 0 & 0 \\ 32 & 0 & 16 \end{bmatrix} \Leftrightarrow$

$\Leftrightarrow H_1 = \begin{bmatrix} -0,6 & 0 & -0,8 \\ 0 & 1 & 0 \\ -0,8 & 0 & 0,6 \end{bmatrix} \mid \cdot A_1^6 \Rightarrow$

[Aplicăm transf^{ă} H la stânga mat A] $\Rightarrow$

$\Rightarrow$ Introducem zerouri

$$\Rightarrow H_1 \cdot A_1 = \begin{bmatrix} -5 & -2 & -5 \\ 0 & 3 & 5 \\ 0 & 4 & 0 \end{bmatrix}$$

not A_2

Obs.: Alg. HH NU mod. el. din stânga $\Rightarrow$ lucrăm cu submatrici ai. să diminuăm calculele și aplicăm primul pas de la HH.

Pas $p = 2$:

$$A_2 = \begin{bmatrix} 3 & 5 \\ 4 & 0 \end{bmatrix}, \quad \tilde{a}_1$$

$$r_2 = \text{semn}(\tilde{a}_{11}) \cdot \|\tilde{a}_1\| = +\sqrt{3^2+4^2} = 5$$

$$v_2 = \begin{bmatrix} 3+5 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}; \quad \|v_2\| = \sqrt{8^2+4^2} = \sqrt{80}$$

$$\Rightarrow H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & I_2 - \frac{2 v_2 v_2^T}{\|v_2\|^2} \end{bmatrix}$$

Pt. a putea „·" cu A_2 3×3

Altfel, $H_2 = I_3 - 2 \cdot \frac{v_2 \cdot v_2^T}{\|v_2\|^2}$; $v_2 = \begin{bmatrix} 0 \\ 8 \\ 4 \end{bmatrix}$

$I_2 - 2 \cdot \frac{v_2 v_2^T}{\|v_2\|^2} = (\ldots)$

$$\Rightarrow H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0{,}6 & -0{,}8 \\ 0 & -0{,}8 & 0{,}6 \end{bmatrix} \cdot A_2 \Rightarrow$$

$$\Rightarrow H_2 \cdot A_2 = \begin{bmatrix} -5 & -2 & -5 \\ 0 & -5 & -3 \\ 0 & 0 & -4 \end{bmatrix} = R \text{ (sup. } \Delta)$$

not A_3

Deci, $R = \begin{bmatrix} -5 & -2 & -5 \\ 0 & -5 & -3 \\ 0 & 0 & -4 \end{bmatrix}$ și $Q = H^T$

$$H = H_2 \cdot H_1 \Rightarrow Q = (H_2 \cdot H_1)^T$$

Morala: A $\rightarrow$ plină (puține zerouri) $\Rightarrow$ HH

$\rightarrow$ goală (multe zerouri) $\Rightarrow$ Givens

Gram - Schmidt

- $A = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix} = [a_1 \; a_2 \; a_3]$ $= [q_1 \; q_2 \; q_3] \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{bmatrix}$

Pasul 1: $r_{11} = \|a_1\| = \sqrt{3^2+0^2+4^2} = 5$

$q_{11} = \frac{a_1}{\|a_1\|} = \frac{1}{5} \cdot \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$

Pasul 2: $r_{12} = \langle q_1, a_2 \rangle = \frac{1}{5} \cdot (-2 \cdot 3 + 3 \cdot 0 + 4 \cdot 4) = 2$

$u_2 = a_2 - r_{12} \cdot q_1 = \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix} - \frac{2}{5} \cdot \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} -16/5 \\ 3 \\ 12/5 \end{bmatrix}$

$\Rightarrow r_{22} = \|u_2\| = 5$

$q_{22} = \frac{u_2}{\|u_2\|} = \begin{bmatrix} -16/25 \\ 3/5 \\ 12/25 \end{bmatrix}$

Pasul 3: $r_{13} = \langle q_1, a_3 \rangle = 3$

$r_{23} = \langle q_2, a_3 \rangle = 3$

$u_3 = a_3 - r_{13} q_1 - r_{13} q_2 = (\ldots) = \begin{bmatrix} 48/25 \\ 16/5 \\ -36/25 \end{bmatrix}$

$\Rightarrow r_{33} = \|u_3\| = 4$

$q_3 = \frac{u_3}{\|u_3\|} = \begin{bmatrix} 12/25 \\ 4/5 \\ -9/25 \end{bmatrix}$

(Elim. Gaussiană)

Eliminare Gaussiană („clasică")

G

$$\bullet\ A = \begin{bmatrix} 1 & 2 & 4 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix};\quad b = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

Obs.: Operațiile pe linii NU modifică sol. sist.

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 1 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 4 \end{array}\right] \xrightarrow[L_3 - 7L_1]{L_2 - 4L_1} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 1 \\ 0 & -3 & -10 & -1 \\ 0 & -6 & -19 & -3 \end{array}\right] \rightarrow$$

$$\xrightarrow{L_3 - 2L_2} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 1 \\ 0 & -3 & -10 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right] \uparrow \text{SST}$$

GPP: Elim. Gaussiană cu pivotare parțială

Se alege un pivot la fiecare pas ($p = 1 : n-1$)

$$|a_{l_{pp}}| = \max_{i=p:n} \{|a_{ip}|\}$$

$$\bullet\ \tilde{A}|\tilde{b} = \left[\begin{array}{ccc|c} \boxed{1} & 2 & 4 & 1 \\ 4 & 5 & 6 & 3 \\ \boxed{7} & 8 & 9 & 4 \end{array}\right] \xleftrightarrow[\text{Pivotare el.}]{L_3 \leftrightarrow L_1} \left[\begin{array}{ccc|c} \boxed{7} & 8 & 9 & 4 \\ \boxed{4} & 5 & 6 & 3 \\ \boxed{1} & 2 & 4 & 1 \end{array}\right] \xrightarrow[L_3 - \frac{1}{7}L_1]{L_2 - \frac{4}{7}L_1}$$

$$\rightarrow \left[\begin{array}{ccc|c} 7 & 8 & 9 & 4 \\ 0 & 3/7 & 6/7 & 5/7 \\ 0 & \boxed{6/7} & 19/7 & 3/7 \end{array}\right] \xleftrightarrow{L_2 \leftrightarrow L_3} (\text{Apoi}) \rightarrow \text{Elim. gauss. clasică}$$

$$\rightarrow \left[\begin{array}{ccc|c} 7 & 8 & 9 & 4 \\ 0 & 6/7 & 19/7 & 3/7 \\ 0 & 0 & -12 & 12 \end{array}\right] \uparrow \text{SST}$$

GPPS: **Elim. Gaussiană cu pivotare parțială cu pivot scalat**

Se alege un pivot la fiecare pas ($p = 1 : n-1$)

$$\frac{|a_{lpp}|}{s_l} = \max_{i=p:n} \left\{ \frac{|a_{ip}|}{s_i} \right\}; \quad s_i = \text{factor de scalare}$$

$$s_i = \max_{j=p:n} \{ |a_{ij}| \}$$

Ex:

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 1 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 4 \end{array}\right]$$

$s_1 = \max\{|1|, |2|, |4|\} = 4$

$s_2 = \max\{|4|, |5|, |6|\} = 6$; Pivotul: $\max\left\{\frac{|1|}{4}, \frac{|4|}{6}, \frac{|7|}{9}\right\} = \frac{7}{9} \Rightarrow$

$s_3 = \max\{|7|, |8|, |9|\} = 9$

$\Rightarrow$ Pivotul $7 \in L_3 \Rightarrow L_1 \leftrightarrow L_3$ + Elim. G clasică $\Rightarrow$

$$\Rightarrow \left[\begin{array}{ccc|c} 7 & 8 & 9 & 4 \\ 0 & 3/7 & 6/7 & 5/7 \\ 0 & 6/7 & 19/7 & 3/7 \end{array}\right]$$

$s_1 = \max\left\{\left|\frac{3}{7}\right|, \left|\frac{6}{7}\right|\right\} = \frac{6}{7}$

$s_2 = \max\left\{\left|\frac{6}{7}\right|, \left|\frac{19}{7}\right|\right\} = \frac{19}{7}$; Pivotul: $\max\left\{\frac{|3/7|}{6/7}, \frac{|6/7|}{19/7}\right\} = \frac{|3/7|}{6/7}$

$\Rightarrow$ Pivotul $\frac{3}{7} \in L_2 \Rightarrow$ Este deja „la locul potrivit"

$\Rightarrow$ (...) $\uparrow$ SST

GPT : Eliminare Gaussiană cu pivotare totală

Se alege un pivot la fiecare pas : $p = 1 : n-1$

$$a_{l_p c_p} = \max_{\substack{p \le i \le n \\ p \le j \le n}} \{ |a_{ij}| \} \Leftrightarrow$$ | Maximul | el. dintr-o submatrice □

X Y Z

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 1 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 4 \end{array}\right] \xleftrightarrow{\substack{L_1 \leftrightarrow L_3 \\ C_1 \leftrightarrow C_3}} \left[\begin{array}{ccc|c} 9 & 8 & 7 & 4 \\ 0 & -1/3 & -2/3 & 1/3 \\ 0 & -14/9 & -19/9 & -7/9 \end{array}\right]$$

+ Ellm. G. clasică $\Leftrightarrow$ $L_2 - \frac{5}{9}L_1 ; L_3 - \frac{4}{9}L_1$

Z X Y

$$\left[\begin{array}{ccc|c} 9 & 7 & 8 & 4 \\ 0 & -19/9 & -14/9 & -7/9 \\ 0 & 0 & 9/57 & 11/19 \end{array}\right] \uparrow \text{SST}$$

$L_3 \leftrightarrow L_2$, $C_3 \leftrightarrow C_2$ $\Leftrightarrow$ (...) $\Leftrightarrow$ Ellm. G. clasică

Trebuie să ținem un vector de permutații, deoarece schimbăm (facem pivot) coloanele, deci schimbăm ordinea necunoșc. (x, y, z)

$A = N - P \quad \| \quad x^{(p+1)} = \underbrace{N^{-1} \cdot P}_{G} \cdot x^{(p)} + \underbrace{N^{-1} \cdot b}_{c} \quad \| \quad G = N^{-1} \cdot P \quad c = N^{-1} \cdot b$

Metode iterative

Jacobi

- $A = \begin{bmatrix} 1 & 2 & 4 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$; $b = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$

Test de convergență - metoda Jacobi:

$A = N - P$, unde $\begin{cases} N = D \\ P = L + U \end{cases}$

mat. de iter. $G_J = N^{-1} \cdot P \Rightarrow G_J = D^{-1} \cdot (L + U)$

$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$; $L = \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ -7 & -8 & 0 \end{bmatrix}$; $U = \begin{bmatrix} 0 & -2 & -4 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix}$

$A = \underbrace{D}_{N} - \underbrace{(L+U)}_{P}$

$D^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/5 & 0 \\ 0 & 0 & 1/9 \end{bmatrix} \Rightarrow G_J = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/5 & 0 \\ 0 & 0 & 1/9 \end{bmatrix}}_{D^{-1}} \cdot \underbrace{\begin{bmatrix} 0 & -2 & -4 \\ -4 & 0 & -6 \\ -7 & -8 & 0 \end{bmatrix}}_{L+U}$

$\Rightarrow G_J = \begin{bmatrix} 0 & -2 & -4 \\ -4/5 & 0 & -6/5 \\ -7/9 & -8/9 & 0 \end{bmatrix}$ - matricea de iterație

Determin val. proprii ale mat. G_J:

$\det(G_J - \lambda I_3) = 0 \Rightarrow \lambda_i \in \{0,97; 1,76; -2,73\}$

$\max |\lambda_i| = 2,73 \geq 1 \Rightarrow$ [J] - NU converge!

Totuși, calc. primii termeni ...: [dependențele $x^{(p+1)} \leftarrow (p)$ se obțin din scrierea liniară a sist. $x^{(p+1)} = G \cdot x^{(p)} + c$]

$$\begin{cases} x_1^{(p+1)} + 2x_2 + 4x_3 = 1 \\ 4x_1 + 5x_2^{(p+1)} + 6x_3 = 3 \\ 7x_1 + 8x_2 + 9x_3^{(p+1)} = 4 \end{cases} \Longleftrightarrow$$

$$\Longleftrightarrow \begin{cases} x_1^{(p+1)} = 1 - 2 \cdot x_2^{(p)} - 4 \cdot x_3^{(p)} \\ x_2^{(p+1)} = \frac{3}{5} - \frac{4}{5} x_1^{(p)} - \frac{6}{5} x_3^{(p)} \\ x_3^{(p+1)} = \frac{4}{9} - \frac{7}{9} x_1^{(p)} - \frac{8}{9} x_2^{(p)} \end{cases}$$

Avantaj: Op. se pot paraleliza

$x_2^{(p+1)}$ ... nu depinde de $x_1^{(p+1)}$ sau $x_2^{(p+1)}$

$X^{(0)} = [0 \ 0 \ 0]^t$ (aprox. inițială)

$p = 0$:
$$\begin{cases} x_1^{(1)} = 1 - 0 - 0 = 1 \\ x_2^{(1)} = \frac{3}{5} - 0 - 0 = 3/5 \\ x_3^{(1)} = \frac{4}{9} - 0 - 0 = 4/9 \end{cases} \Rightarrow X^{(1)} = [1 \ 3/5 \ 4/9]$$

$p = 1$:
$$\begin{cases} x_1^{(2)} = 1 - 2 \cdot \frac{3}{5} - 4 \cdot \frac{4}{9} = 1 - \frac{6}{5} - \frac{16}{9} \\ x_2^{(2)} = \frac{3}{5} - \frac{4}{5} \cdot 1 - \frac{6}{5} \cdot \frac{4}{9} = \frac{3}{5} - \frac{4}{5} - \frac{24}{45} \\ x_3^{(2)} = \frac{4}{9} - \frac{7}{9} \cdot 1 - \frac{8}{9} \cdot \frac{3}{5} = \frac{4}{9} - \frac{7}{9} - \frac{24}{45} \end{cases} \Rightarrow$$

$$\Rightarrow X^{(2)} = [-1.97, \ -0.73, \ -0.86]^t$$

$p = 2$: (...)

În concluzie $\rightarrow$ NU converge (la sol. exactă)

$A = N - P$ (la model general)

Jacobi: $A = \underbrace{D}_{N} - \underbrace{[L + U]}_{P}$

Gauss - Seidel

GS: $A = \underbrace{[D - L]}_{N} - \underbrace{U}_{P}$

- $A = \begin{bmatrix} 1 & 2 & 4 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$; $b = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$

$$G = N^{-1} \cdot P$$
$$c = N^{-1} \cdot b$$

$$x^{(p+1)} = G \cdot x^{(p)} + c$$

Test de convergență - metoda Gauss-Seidel:

$A = N - P$, unde $\begin{cases} N = D - L \\ P = U \end{cases}$

$G_{GS} = N^{-1} \cdot P \Rightarrow G_{GS} = (D - L)^{-1} \cdot U$

$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$; $L = \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ -7 & -8 & 0 \end{bmatrix}$; $U = \begin{bmatrix} 0 & -2 & -4 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix}$

$$G_{GS} = \underbrace{(D-L)}_{N}^{-1} \cdot U = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 5 & 0 \\ 7 & 8 & 9 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 0 & -2 & -4 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix} \overset{(...)}{\Leftrightarrow}$$

$$\Leftrightarrow G_{GS} = \begin{bmatrix} 0 & -2 & -4 \\ 0 & 8/5 & 2 \\ 0 & 2/15 & 4/3 \end{bmatrix}$$

$\det(G_{GS} - \lambda I_3) = 0 \Rightarrow \lambda_i \in \{0, \frac{14}{15}, 2\}$

$\max|\lambda_i| = 2 \not< 1 \Rightarrow$ GS - NU converge

Totuși, calc. primii termeni (...):

$$\begin{cases} x_1^{(p+1)} + 2x_2 + 4x_3 = 1 \\ 4x_1^{(p+1)} + 5x_2^{(p+1)} + 6x_3 = 3 \\ 7x_1^{(p+1)} + 8x_2^{(p+1)} + 9x_3^{(p+1)} = 4 \end{cases} \iff$$

$$\iff \begin{cases} x_1^{(p+1)} = 1 - 2x_2^{(p)} - 4x_3^{(p)} \\ x_2^{(p+1)} = \frac{3}{5} - \frac{4}{5}x_1^{(p+1)} - \frac{6}{5}x_3^{(p)} \\ x_3^{(p+1)} = \frac{4}{9} - \frac{7}{9}x_1^{(p+1)} - \frac{8}{9}x_2^{(p+1)} \end{cases}$$

Detaliu:
Op. nu se pot //

Avantaj:
pt. $x_2^{(p+1)}$ se fol. $x_1^{(p+1)}$ ⇒ Informația "mai recentă"; actual

$x^{(0)} = [0\ 0\ 0]'$

$p = 0 \Rightarrow (\dots) \Rightarrow x^{(1)} = [1,\ -0.2,\ -0.15]'$

$p = 1 \Rightarrow (\dots) \Rightarrow x^{(2)} = [\dots]'$

În concluzie → MJ converge (la sol. exactă)

SOR - Metoda supraralaxării → GS + improve

- $A = \begin{bmatrix} 1 & 2 & 4 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$; $b = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$

$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$; $L = \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ -7 & -8 & 0 \end{bmatrix}$; $U = \begin{bmatrix} 0 & -2 & -4 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix}$

$\omega \in (0,2)$:

Pt. $\omega = 1 \Rightarrow$ SOR = GS

Consider: $\boxed{\omega = \frac{3}{2}}$ (1,5)

- $N(\omega) = \frac{1}{\omega} \cdot D - L \Rightarrow N = \frac{2}{3} \cdot D - L \Rightarrow$

$\Rightarrow N = \begin{bmatrix} 2/3 & 0 & 0 \\ 0 & 10/3 & 0 \\ 0 & 0 & 6 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ -7 & -8 & 0 \end{bmatrix} \Rightarrow$

$\Rightarrow N = \begin{bmatrix} 2/3 & 0 & 0 \\ 4 & 10/3 & 0 \\ 7 & 8 & 6 \end{bmatrix}$

- $P(\omega) = \frac{1-\omega}{\omega} \cdot D + U \Rightarrow P = \frac{-1/3}{\frac{1}{-\frac{3}{2}}} \cdot D + U \Rightarrow$

$\Rightarrow P = \begin{bmatrix} -1/3 & 0 & 0 \\ 0 & -5/3 & 0 \\ 0 & 0 & -3 \end{bmatrix} + \begin{bmatrix} 0 & -2 & -4 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow$

$\Rightarrow P = \begin{bmatrix} -1/3 & -2 & -4 \\ 0 & -5/3 & -6 \\ 0 & 0 & -3 \end{bmatrix}$

$$G_{SOR}(\omega) = N^{-1}(\omega) \cdot P(\omega) \quad (\text{pt } \omega = \frac{3}{2})$$

$$\Rightarrow G_{SOR} = \begin{bmatrix} 2/3 & 0 & 0 \\ 4 & 10/3 & 0 \\ 7 & 8 & 6 \end{bmatrix}^{-1} \cdot \begin{bmatrix} -1/3 & -2 & -4 \\ 0 & -5/3 & -6 \\ 0 & 0 & -3 \end{bmatrix}$$

(...)

Val. proprii: $\lambda_i \in \{-0.14,\ 0.57,\ 1.46\}$

$\max|\lambda_i| = 1,46 \geq 1 \Rightarrow$ **SOR** - NU converge

Totuși, calc. primii termeni (...):

GS:
$$\begin{cases} x_1^{(\phi+1)} = 1 - 2x_2^{(\phi)} - 4x_3^{(\phi)} \\ x_2^{(\phi+1)} = \frac{3}{5} - \frac{4}{5}x_1^{(\phi+1)} - \frac{6}{5}x_3^{(\phi)} \\ x_3^{(\phi+1)} = \frac{4}{9} - \frac{7}{9}x_1^{(\phi+1)} - \frac{8}{9}x_2^{(\phi+1)} \end{cases}$$

aprox. inițiale: $x^{(0)} = [0\ 0\ 0]'$

$\phi = 0 \Rightarrow (\overset{GS}{\ldots}) \Rightarrow x^{(1)} = [1,\ -0.2,\ -0.15]'$

Corectăm sol. găsită cu GS, astfel:

- $x_1^{(1)} = \omega \cdot x_1^{(1)} + (1-\omega) \cdot x_1^{(0)} \Leftrightarrow$

$\Leftrightarrow x_1^{(1)} = \frac{3}{2} \cdot 1 + \left(1 - \frac{3}{2}\right) \cdot 0 \Rightarrow \boxed{x_1^{(1)} = \frac{3}{2}}$

- $x_2^{(1)} = \frac{3}{2} \cdot (-0,2) - \frac{1}{2} \cdot 0 \Rightarrow \boxed{x_2^{(1)} = -\frac{3}{10}}$

- $x_3^{(1)} = \frac{3}{2} \cdot (-0,15) - \frac{1}{2} \cdot 0 \Rightarrow \boxed{x_3^{(1)} = \frac{9}{40}}$

$\Rightarrow \boxed{x^{(1)}} = [3/2 \quad -3/10 \quad 9/40]'$

$(\phi = 1 (\ldots))$ În concluzie → NU converge (la sol. exactă)

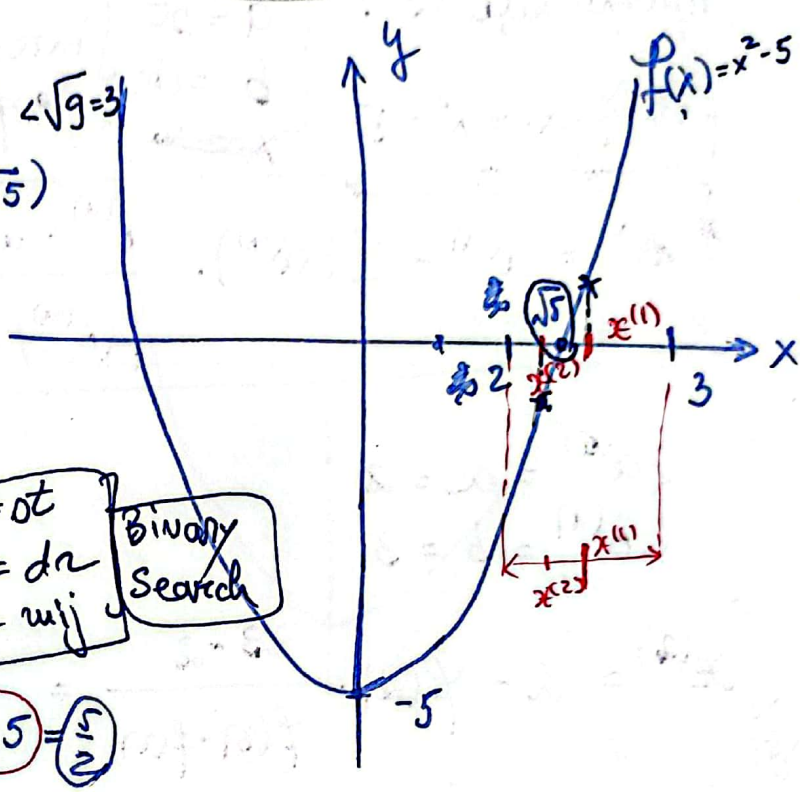
Ec. neliniare

Metoda bisecției

$f(x) = x^2 - 5$ (aproximați $\sqrt{5}$)

$2 = \sqrt{4} < \sqrt{5} < \sqrt{9} = 3$

$a^{(1)} = \{a = 2$
$b^{(1)} = \{b = 3$ } intervalul inițial



$$x^{(1)} = a^{(1)} + \frac{b^{(1)} - a^{(1)}}{2}$$

a = ot
b = dr
x = mij
Binary Search

$\Rightarrow x^{(1)} = 2 + \frac{3-2}{2} = 2 + \frac{1}{2} = 2{,}5 = \frac{5}{2}$

$f(a^{(1)}) \cdot f(x^{(1)}) < 0 \Rightarrow b^{(2)} = x^{(1)}$ și $a^{(2)} = a^{(1)}$ (Rămâne)

$f(a^{(1)})$: $a^{(1)}$ — <0 ; $f(x^{(1)})$: $x^{(1)}$ — >0

$x^{(2)} = a^{(2)} + \frac{b^{(2)} - a^{(2)}}{2} \Rightarrow x^{(2)} = 2 + \frac{\frac{5}{2} - 2}{2} = 2 + \frac{1}{4} = \frac{9}{4}$

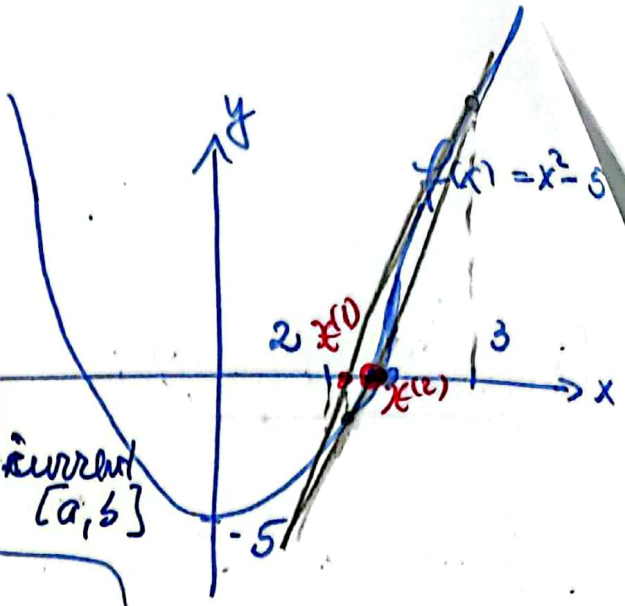
$= 2{,}25$

$f(a^{(2)}) \cdot f(x^{(2)}) > 0 \Rightarrow a^{(3)} = x^{(2)}$ și $b^{(3)} = b^{(2)}$ (Rămâne)

$f(a^{(2)}) = f(a^{(1)}) < 0$; $f(x^{(2)}) < 0$

$x^{(3)} = (\ldots)$

Metoda secantei



Aproximați $\sqrt{5}$:

fie $f(x) = x^2 - 5$

$\left[\begin{array}{l} a = st \\ b = dr \end{array}\right\}$ intervalul la pasul curent $[a, b]$

$$x^{(n)} = a^{(n)} - f(a^{(n)}) \cdot \frac{b^{(n)} - a^{(n)}}{f(b^{(n)}) - f(a^{(n)})}$$

$$\begin{cases} a^{(1)} = a = 2 \\ b^{(1)} = b = 3 \end{cases}$$

Pas 1

$$x^{(1)} = 2 - f(2) \cdot \frac{3-2}{f(3)-f(2)} = 2 + \frac{1}{4+1} = 2 + \frac{1}{5} = \frac{11}{5} \ (2,2)$$

$f(a^{(1)}) \cdot f(x^{(1)}) > 0 \Rightarrow$ $a^{(2)} = x^{(1)}$ și $b^{(2)} = b^{(1)}$

($a^{(1)} < 0$, $f(x^{(1)}) < 0$)

(Rămâne fixat)

$$x^{(2)} = \frac{11}{5} - f\left(\frac{11}{5}\right) \cdot \frac{3 - \frac{11}{5}}{f(3) - f\left(\frac{11}{5}\right)} = \frac{11}{5} + \frac{121}{25} \cdot \frac{\frac{4}{5}}{4 + \frac{121}{25}} \Leftrightarrow$$

$$\Leftrightarrow x^{(2)} = \frac{11}{5} + \frac{4}{25} \cdot \frac{4}{29} \cdot \frac{25}{104} \Leftrightarrow x^{(2)} = \frac{29}{13} \ (\approx 2,2308)$$

Pas 2

$f(a^{(2)}) \cdot f(x^{(2)}) > 0 \Rightarrow$ $a^{(3)} = x^{(2)}$

($f(x^{(1)}) < 0$, $f(x^{(2)}) < 0$)

și $b^{(3)} = b^{(2)}$ (Rămâne fixat)

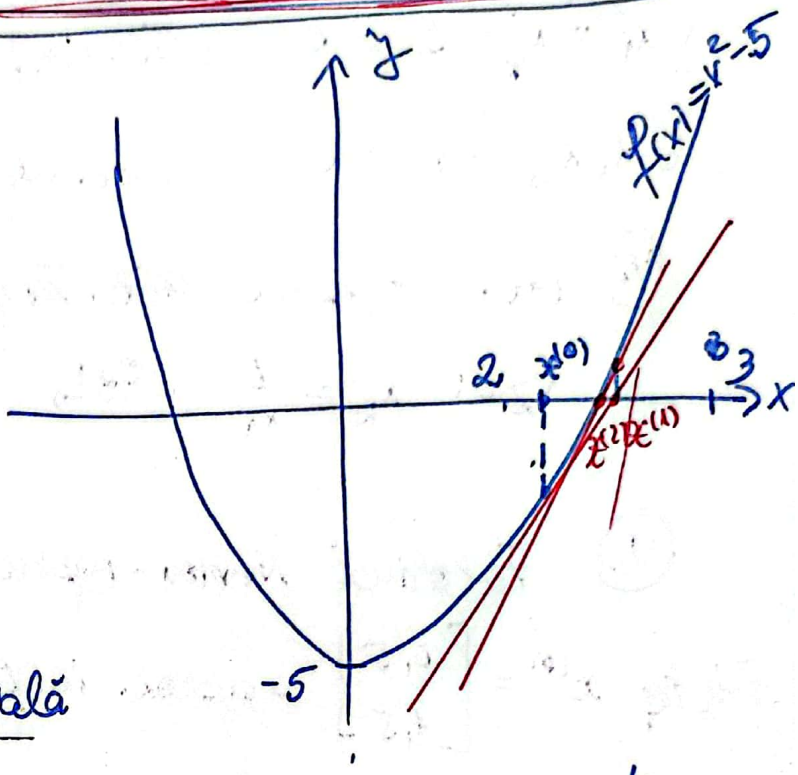
(...)

Metoda tangentei (Metoda Newton - Raphson)

Aproximați $\sqrt{5}$:

Fie $f(x) = x^2 - 5$

$$x^{(n+1)} = x^{(n)} - \frac{f(x^{(n)})}{f'(x^{(n)})}$$



Fie $x^{(0)} = 2$ - aprox. inițială

$n=0$: $x^{(1)} = x^{(0)} - \frac{f(x^{(0)})}{f'(x^{(0)})} \Leftrightarrow x^{(1)} = 2 - \frac{-1}{4} = 2 + \frac{1}{4} =$

$\Leftrightarrow x^{(1)} = \frac{9}{4}$ $(\simeq 2{,}25)$

$n=1$: $x^{(2)} = x^{(1)} - \frac{f(x^{(1)})}{f'(x^{(1)})} \Leftrightarrow x^{(2)} = \frac{9}{4} - \frac{\frac{81}{16} - 5}{\frac{9}{2}} \Leftrightarrow$

$\Leftrightarrow x^{(2)} = \frac{161}{72}$ $(\simeq 2{,}2361)$

Sisteme neliniare - Metoda Newton-Raphson

$$\begin{cases} x_1^2 + x_2^2 - 3 = 0 & (\text{Ec. cerc}) \\ x_1 x_2 - 1 = 0 & (\text{Ec. hiperbolă}) \end{cases}$$

$$\boxed{\begin{array}{l} J(x^{(k)}) \cdot \Delta x^{(k)} = -F(x^{(k)}) \\ x^{(k+1)} = \Delta x^{(k)} + x^{(k)} \end{array}}$$

(V1) Rez. o ec cu Met. Tangentei, astfel:

Ec 2: $x_2 = \frac{1}{x_1} \xrightarrow{\text{Ec 1}} x_1^2 + \frac{1}{x_1^2} - 3 = 0$, apoi Met. Tangentei

(V2) Folosind Newton-Raphson - pt. sisteme neliniare

Fie $x^{(0)} = \begin{bmatrix} 0,5 \\ 1,5 \end{bmatrix}$ - aprox. inițială

$$\begin{cases} f_1(x_1, x_2) = x_1^2 + x_2^2 - 3 \\ f_2(x_1, x_2) = x_1 x_2 - 1 \end{cases} ; \quad F = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} ; \quad J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 & 2x_2 \\ x_2 & x_1 \end{bmatrix}$$

Pasul 1: $J(x^{(0)}) \cdot \Delta x^{(0)} = -F(x^{(0)}) \Leftrightarrow$

$$\Leftrightarrow \begin{bmatrix} 2 \cdot 0,5 & 2 \cdot 1,5 \\ 1,5 & 0,5 \end{bmatrix} \cdot \Delta x^{(0)} = -\begin{bmatrix} 0,5^2 + 1,5^2 - 3 \\ 0,5 \cdot 1,5 - 1 \end{bmatrix} \Leftrightarrow$$

$\Leftrightarrow$ Rez. sist. liniar (cu nec. $\Delta x^{(0)}$) $\uparrow$ SST $\Rightarrow \Delta x^{(0)} = \begin{bmatrix} 0,125 \\ 0,125 \end{bmatrix}$

$$\Rightarrow x^{(1)} = \Delta x^{(0)} + x^{(0)} \Rightarrow \boxed{x^{(1)} = \begin{bmatrix} 0,625 \\ 1,625 \end{bmatrix}}$$

Pasul 2: $J(x^{(1)}) \cdot \Delta x^{(1)} = -F(x^{(1)}) \Leftrightarrow (\ldots) \Leftrightarrow$

$$\Leftrightarrow \boxed{x^{(2)} = \begin{bmatrix} 0,61806 \\ 1,61806 \end{bmatrix}}$$

Obs.: Soluția exactă: $x^{(*)} = \begin{bmatrix} 0,618\ldots \\ 1,618\ldots \end{bmatrix}$